

1 Additional Simulation Results - ALYZ

In the following subsections additional results to Puchhammer et al. (2025) are presented. The methods are analyzed for the remaining three main scenarios for the covariance structure based on Agostinelli et al. (2015):

3. $N = 2$ unbalanced groups ($p = 10, n_1 = 100, n_2 = 50$),
4. $N = 2$ balanced groups with increasing singularity issues ($p = 20, n_1 = n_2 = 30$) and
5. high-dimensional $N = 2$ balanced groups ($p = 60, n_1 = n_2 = 40$).

For each setting, the performance of parameter estimation compared to competing methods is visualized as well as the correctness of flagging outlying cells. Moreover, for each scenario a table with the number of repetitions considered in the figures for cellMCD and cellGMM is given. They can deviate from the default number of 100 due to the restriction of the cellMCD regarding the number of marginal outliers and computational issues for the cellGMM in higher dimensional data ($p = 40, 60$).

1.1 Scenario 3 - Unbalanced Scenario

γ_{cell}	$\mu = 0$		$\mu = \text{varying}$	
	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$
2	100	100	100	100
6	100	100	99	100
10	100	100	99	100

Table 1: Number of successful replications for cellMCD in scenario 3 for ALYZ covariance structure.

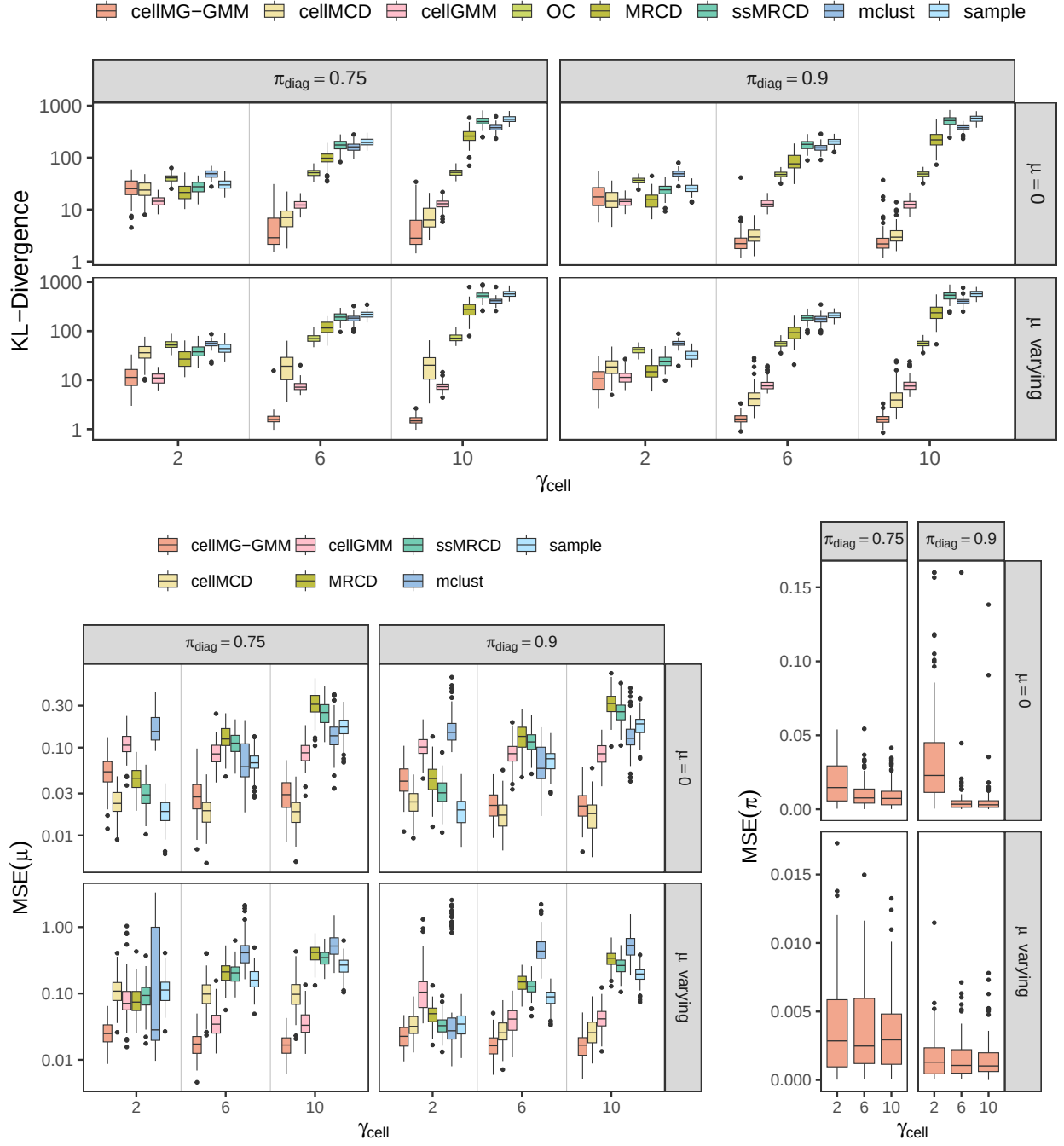


Figure 1: Parameter estimates for scenario 3, the unbalanced groups ($N = 2, p = 10, n_1 = 100, n_2 = 50$) with ALYZ structured covariance matrices. Top panel: KL-divergence of the covariance estimates. Bottom left panel: MSE of the mean estimation. Bottom right panel: MSE of the mixture probabilities π .

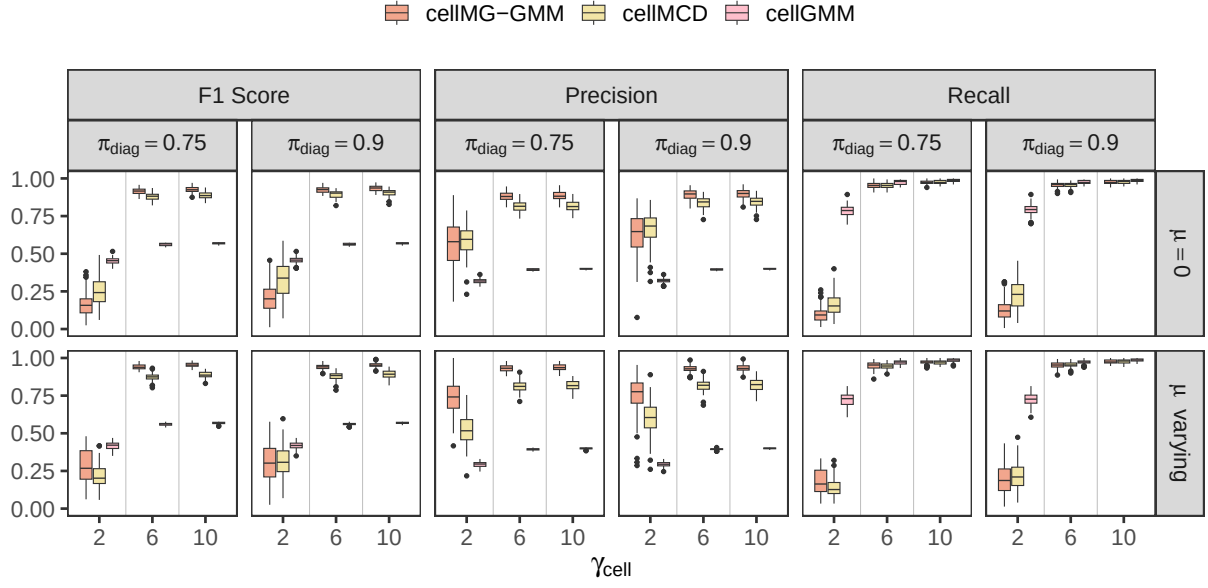


Figure 2: Performance of cellwise outlier detection in scenario 3, unbalanced groups ($N = 2, p = 10, n_1 = 100, n_2 = 50$) with ALYZ structured covariances evaluated by precision, recall and F1-score.

1.2 Scenario 4 - Increasing Singularity Issues

γ_{cell}	$\mu = 0$		$\mu = \text{varying}$	
	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$
2	92	88	92	85
6	81	83	82	82
10	79	82	81	82

Table 2: Number of successful replications for cellMCD in scenario 4 for ALYZ covariance structure.

γ_{cell}	$\mu = 0$		$\mu = \text{varying}$	
	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$	$\pi_{diag} = 0.75$	$\pi_{diag} = 0.9$
2	6	5	8	9
6	20	20	16	13
10	13	18	18	17

Table 3: Number of successful replications for cellGMM in scenario 4 for ALYZ covariance structure.

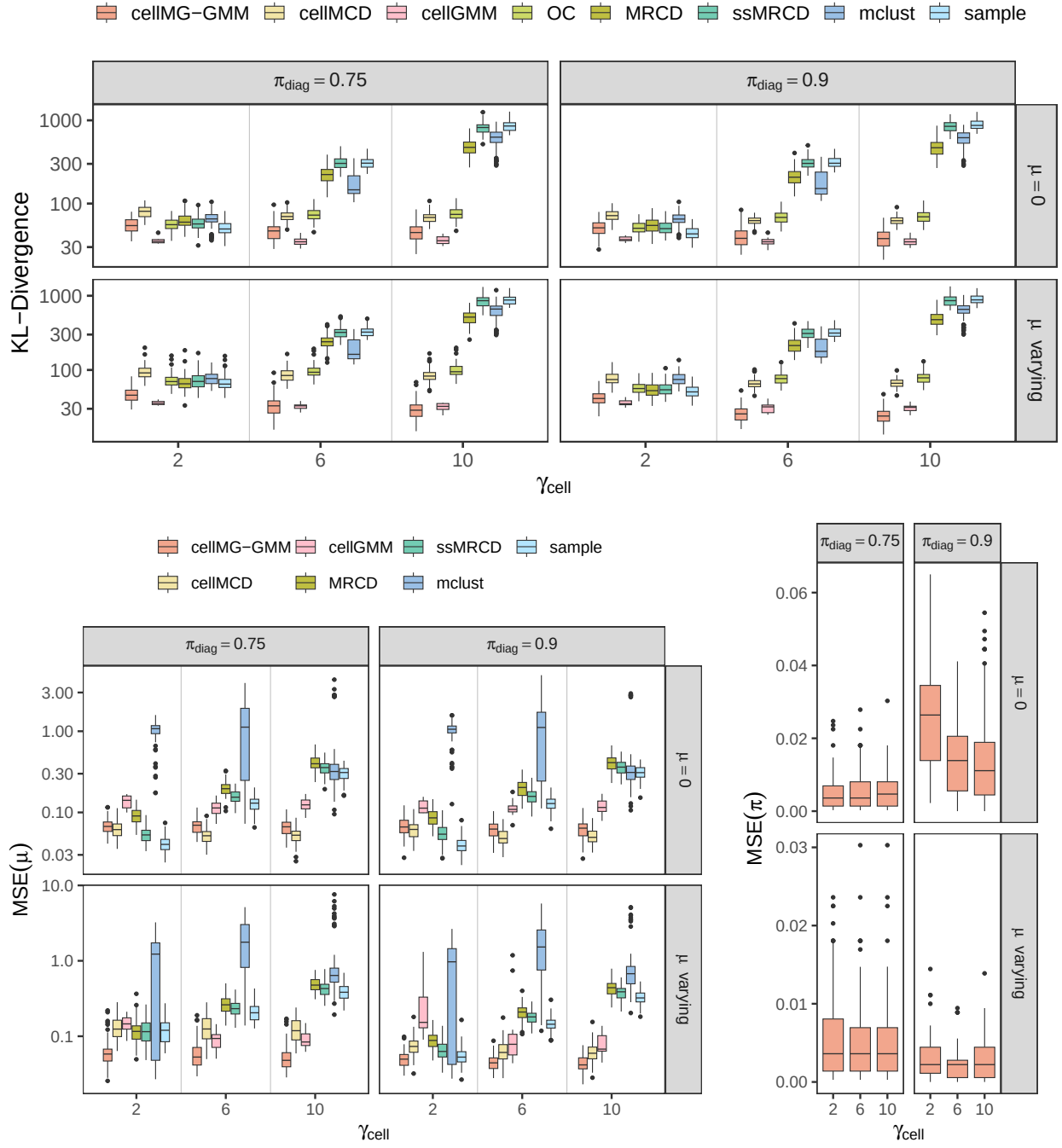


Figure 3: Parameter estimates for scenario 4 ($N = 2, p = 30, n_1 = n_2 = 40$) with ALYZ structured covariance matrices. Top panel: KL-divergence of the covariance estimates. Bottom left panel: MSE of the mean estimation. Bottom right panel: MSE of the mixture probabilities π .

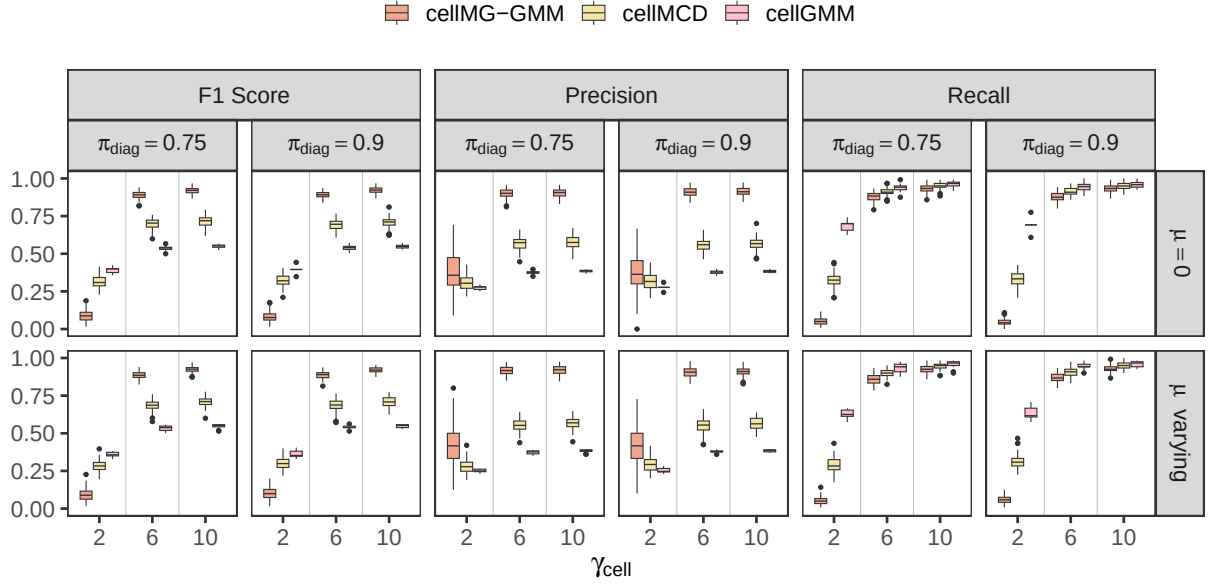


Figure 4: Performance of cellwise outlier detection in scenario 4 ($N = 2, p = 30, n_1 = n_2 = 40$) with ALYZ structured covariances evaluated by precision, recall and F1-score.

1.3 Scenario 5 - High-Dimensional

No calculations of cellMCD and cellGMM were successful and are thus not included.

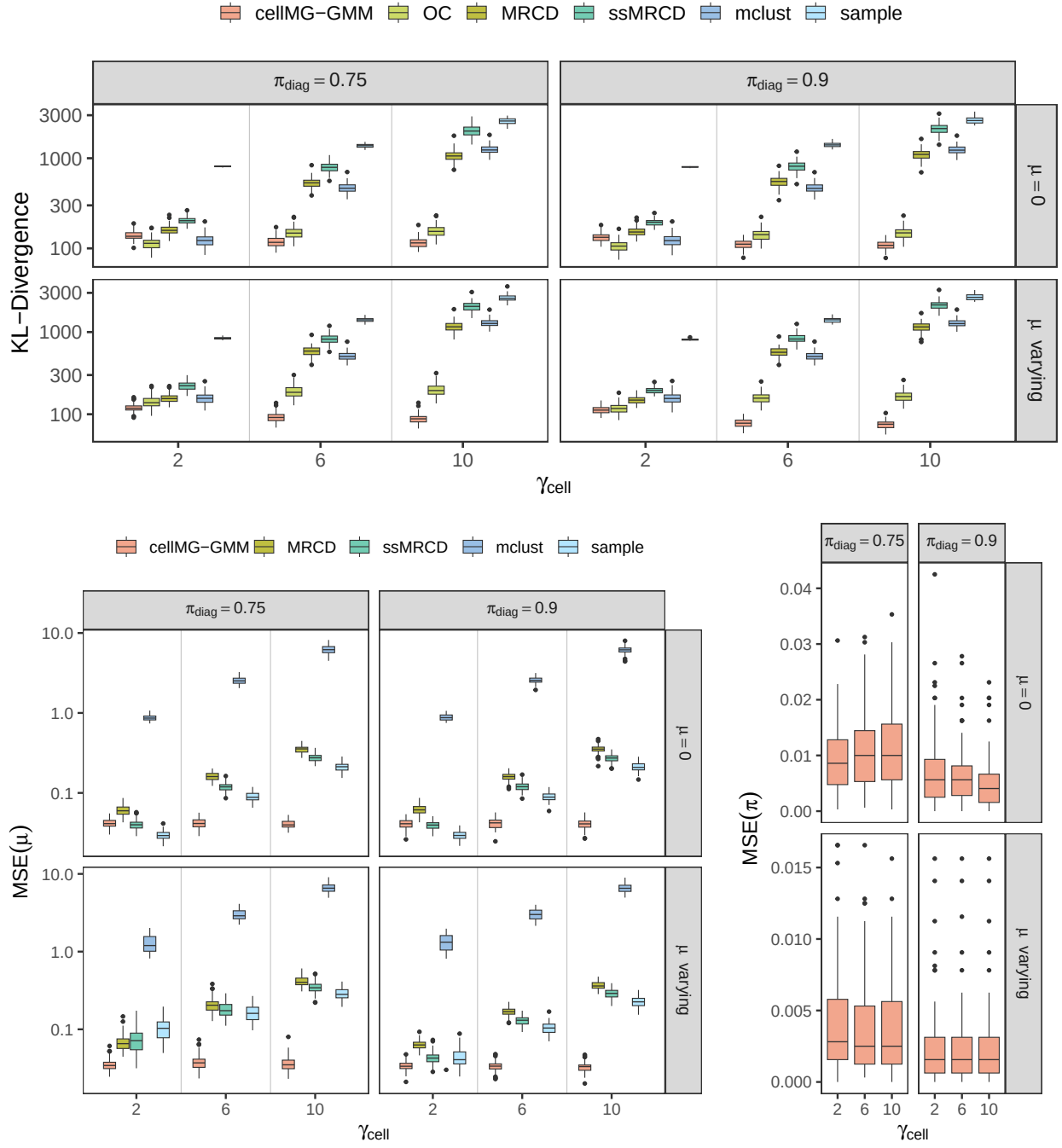


Figure 5: Parameter estimates for scenario 5 ($N = 2, p = 60, n_1 = n_2 = 40$) with ALYZ structured covariance matrices. Top panel: KL-divergence of the covariance estimates. Bottom left panel: MSE of the mean estimation. Bottom right panel: MSE of the mixture probabilities π .

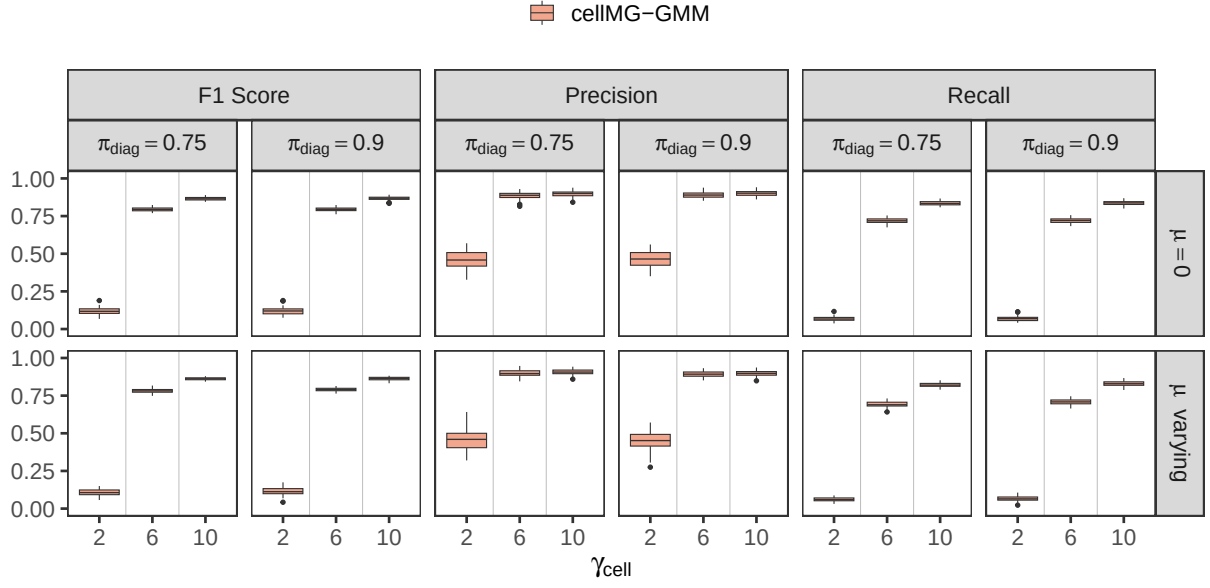


Figure 6: Performance of cellwise outlier detection in scenario 5 ($N = 2, p = 60, n_1 = n_2 = 40$) with ALYZ structured covariances evaluated by precision, recall and F1-score.

References

- Agostinelli, C., Leung, A., Yohai, V. J., and Zamar, R. H. (2015). Robust estimation of multivariate location and scatter in the presence of cellwise and casewise contamination. *Test*, 24:441–461.
- Puchhammer, P., Wilms, I., and Filzmoser, P. (2025). Outlier-robust multi-group gaussian mixture modeling with flexible group reassignment. [arXiv preprint arXiv:2504.02547](https://arxiv.org/abs/2504.02547).